

Abhay Sastha S

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science Engineering

Vellore, TN

Expected July 2026

Indian Institute of Technology, Madras

Bachelor of Science in Data Science and Applications

Chennai, TN

Expected 2026

Experience

MyFi TIFIN India

Bangalore, KA

Machine Learning Intern

December 2025 - Present

- Engineered the frontend and the backend setup along with the integration of ASR and TTS using a custom vllm-omni server optimized for latency and performance to serve Qwen3-tts custom voice. ([here](#))
- Managed and improved Multi-Agent Orchestration setup for financial chatbot, also implemented new Math agent to remove dependency on LLM maths.
- Built an evaluation framework and test bench for agentic-chatbots deployed on Motilal, Axis, SBI and MyFi

Machine Learning Intern

April - August 2025

- Developed a hierarchical graph RAG with attribute extraction and high-order nodes for financial documents,
- Finetuned LLMs, embedding and reranker models over various curated datasets using techniques like Instruction tuning, DPO, LORA, QLORA, negative mining, and more.
- Conducted knowledge distillation from Sarvam-M to Qwen-14B for improved indic capabilities using Response-Matching, then Qwen-14B to 3B using copy-weight initialization for inference time reduction,
- Redesigned the RAG system using a Tip-of-Tongue approach to improve Success@10 from 35% to 85% on a manually curated golden label dataset,
- Refactored the inference pipeline using Golang, decreasing latency from 250ms to 160ms and improving throughput from 10 to 60 requests per second (RPS),
- Implemented guardrails using a custom architecture with finetuned sub-100M Llama models,
- Created a multimodal RAG system package that used Qwen Omni and VLM, along with specially designed table, chart, and image extraction systems using PyOCR and R-CNN FPN, implemented a specialized xml/md: .pang format for custom chunking and visualization of ingested documents,
- Deployed systems developed during this period on MyFi, SBI Mutual Assist WhatsApp bot and DSP chat bot.

Cre-AID Labs

Bangalore, KA

Project Lead Intern

December 2023

- Prototyped and tested an EMG Biofeedback device at 98% lower cost than leading market alternatives.
- Integrated a Live Voice Detection and Actuation Module while improving the app's UI/UX.

Research and Development Intern

July 2023

- Deployed Real-time Voice Detection and Actuation Module to run on a low-resource device with hotwords,
- Built a hand-size measurement system using PyTorch, Detectron, and U2Net, which reduces time by 3/4ths.
- Established framework for crack and abnormality detection, reducing quality check time by 86% (13 minutes)
- Designed a custom GATT-based Bluetooth protocol for the App to communicate with their devices.

Research

- WALO: A Hybrid Whale And Ant Lion Optimization Algorithm for Robust Global Optimization. Evolutionary Intelligence, Springer Nature. ([here](#))
- 4 to 7: A hybrid deep learning model for detection of keratoconus with and without feature selection using EfficientnetV2 (awaiting conference)

Projects

Charto

Value extraction in JSON for unlabelled data points in bar, line, and scattered charts. Converted the Caffe2 model to work on a manual PyTorch Detectron2 implementation of Faster RCNN with FPN. Trained the model on the PlotQA dataset and increased accuracy by adding systematic detections and extractions using PyOCR. Achieved 80-85% accuracy, comparable to SOTA methods such as LlamaParse. [[here](#)]

DI-GNN

Implemented various SOTA methods like MedusaGraph and EquiBind to understand existing methodology. Based on the gaps, devised and implemented a PoC of DI-GNN, an approach that integrates dynamic interaction graphs and RL-guided Pose Refinement for accurate and generalizable solution for drug discovery. [[here](#)]

CoT GRPO LLM

Fine-tuned Qwen2.5-3B-Instruct model using a GRPO-based reinforcement learning framework with custom reward functions to enforce numerical accuracy and strict output formatting. Integrated LoRA adapters with unsloth's FastLanguageModel and vllm for efficient inference. [[here](#)]

Coeus

A Natural Language to SQL converter workflow based on a Vector and Graph RAG system with LLM. Solving issues like hallucination, token size limit, data privacy, and user intent. [[here](#)]

Trackem

Constructed a cross-camera multi-feed human tracking system using YOLO and Re-ID with optimizations for CUDA support. [[here](#)]

Additional Coursework

1. CS6350 - Computer Vision by VPLAB, IIT Madras
2. FCSE753 - Automatic Speech Recognition by Preethi Jyothi, IIT Bombay

Workshops & Certifications

1. Presented poster on OMNIFLUX - IndoML 2025, BITS Hyderabad ([here](#))
2. Foundations in Data Science - IIT Madras
3. Genomic Data Analysis on AWS Omics - IBAB, Bangalore
4. Fine-tuning and Evaluation - DeepLearning.ai
5. Quantization in Depth - HuggingFace, DeepLearning.ai

Activities

VIT Vellore SAS Lab

Student Researcher

Vellore, TN

August 2024 - Present

Worked under Prof. Prabhujit Mohapatra, faculty from the School of Sciences, specializing in Metaheuristic Algorithms for image thresholding. Collaboration resulted in 2 publications: WALO and OBL - CPSOGSA.

Turn It Online (TIO)

Founder,

Vellore, TN

March 2023 - Jun 2024

1. Providing a one-stop solution to businesses trying to establish an online presence.
2. Catered to clients across India, currently working and providing support for 2 other websites.

Skills

Libraries - PyTorch, OpenCV, Keras, YOLO, MediaPipe, NLTK, Scikit, Pandas, Matplotlib, NumPy,
Languages/Scripts - C, C++, Java, Python, Dart, R, HTML, CSS, JS, MySQL
Software - Pinecone, ElasticSearch, MATLAB, Docker, GitHub, HuggingFace, AWS, Microsoft Office Suite, Canva, Figma, Adobe suite
Languages - English, Hindi, Tamil, Malayalam, Sanskrit